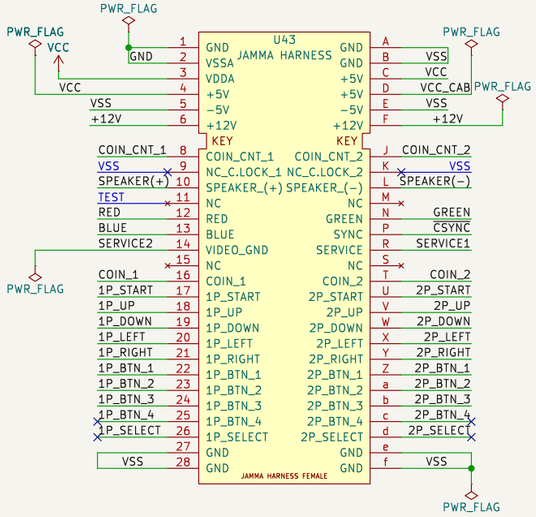
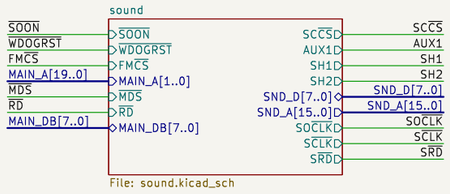
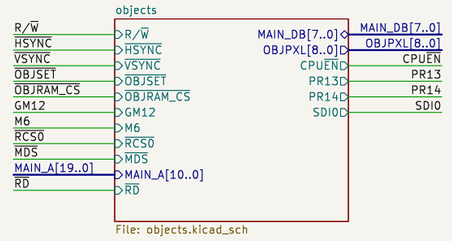
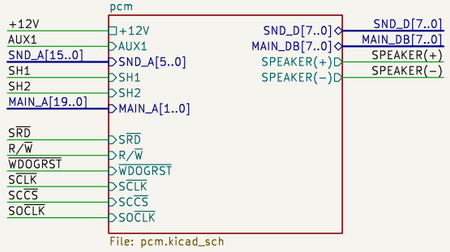
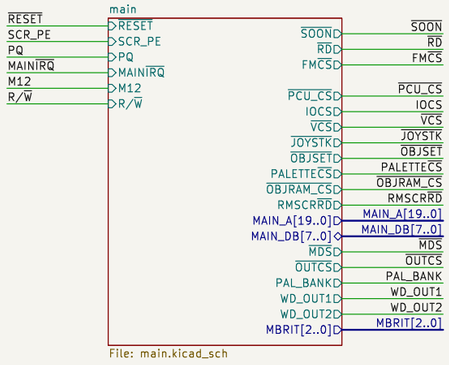
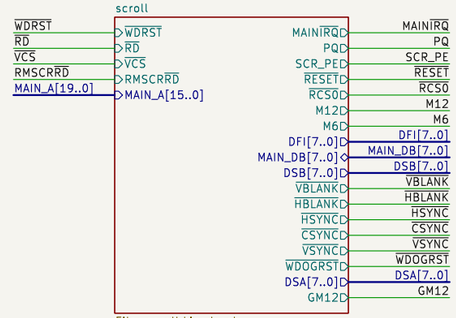


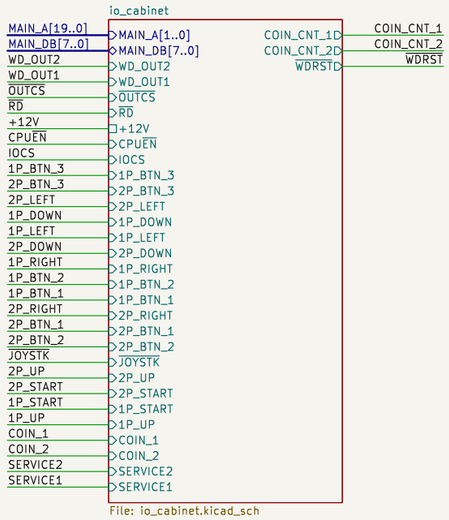


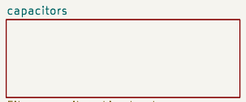


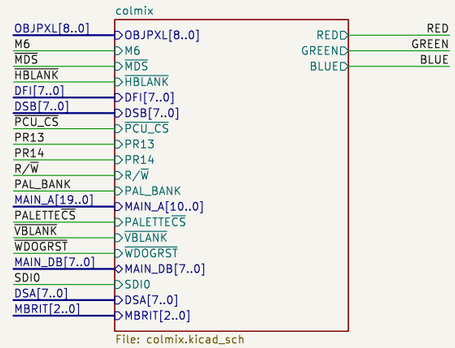












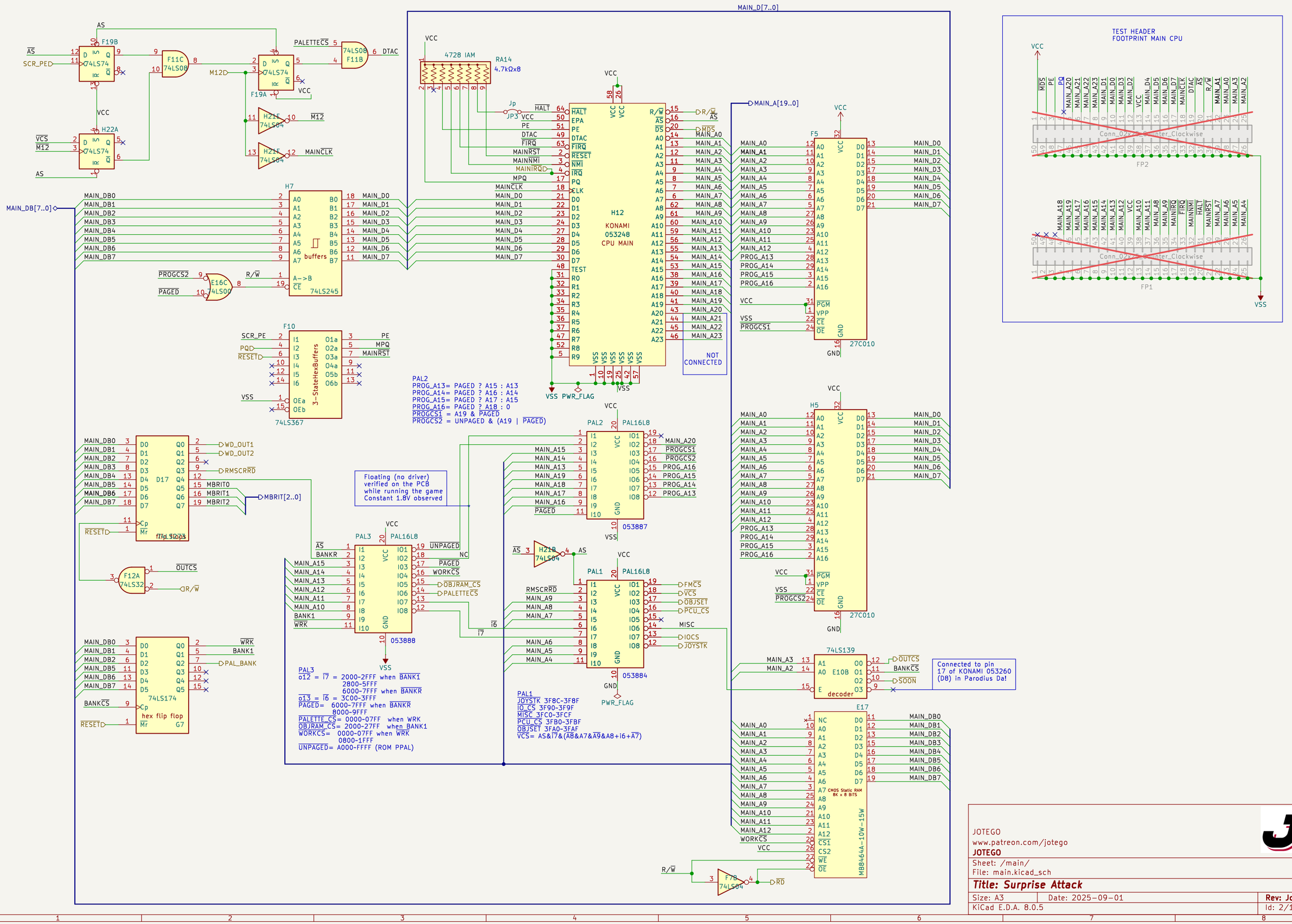
PWB352020B

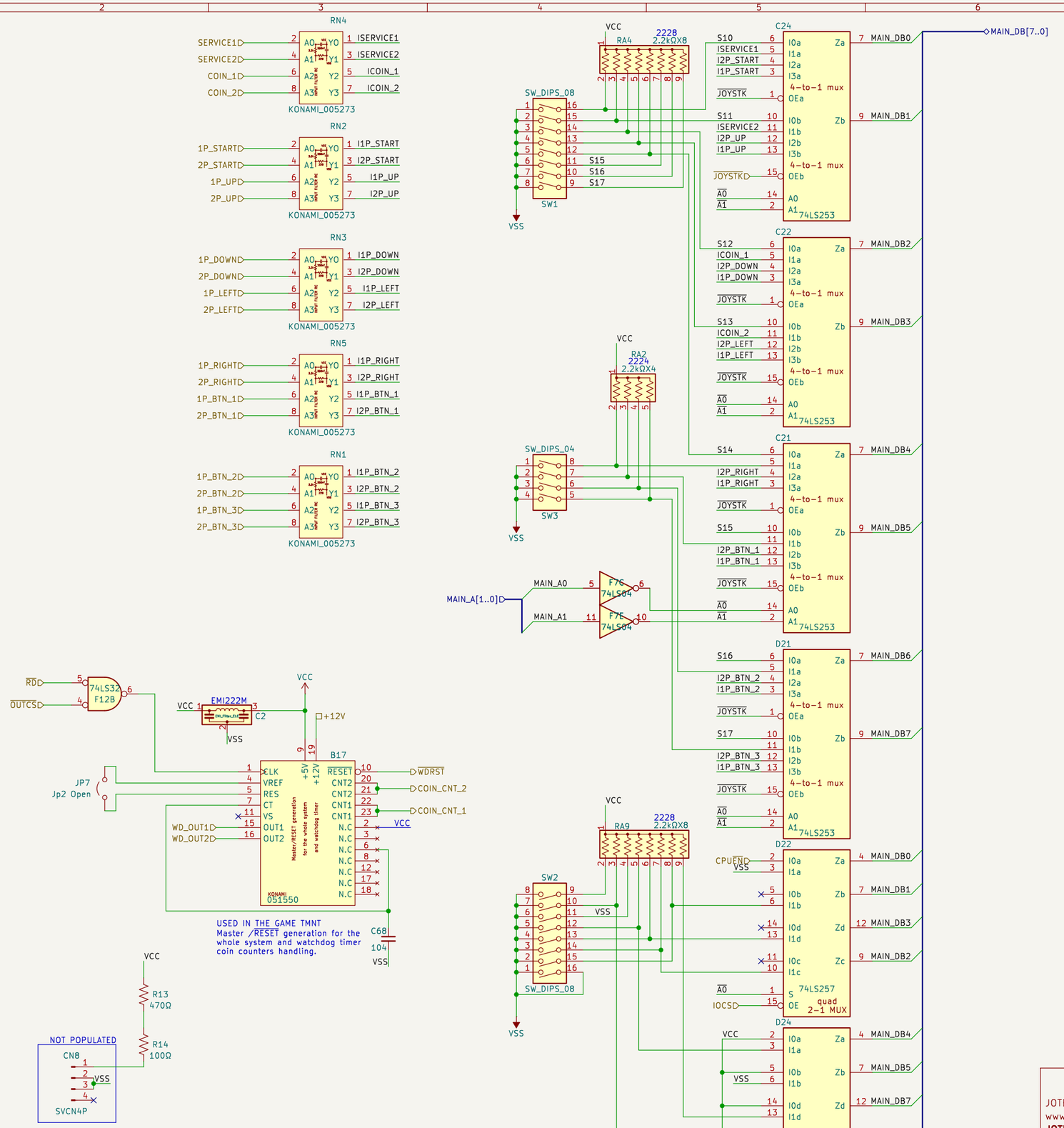
JOTEGO
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Sheet: /
File: surprise_attack.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 1/10


MAIN_D[7..0]
MAIN_A[19..0]
MAIN_DB[7..0]
F19B
F19A
F11C
74LS08
F11B
H22A
M12
AS
SCR_PED
VCS
PALETTECS
DTAC
MAINCLK
H7
buffers
74LS245
PROGCS2
PAGED
R/W
E16C
74LS00
F10
3-StateHexBuffers
74LS367
SCR_PE
PQ
RESET
PE
MPQ
MAINRST
VSS
RA14
4.7kΩx8
4728 IAM
Jp
JP3
HALT
PE
DTAC
FIRQ
MAINRST
MAINNMI
MAINIRQ
MPQ
MAINCLK
H12
KONAMI
053248
CPU MAIN
NOT
CONNECTED
VSS PWR_FLAG
F5
H5
27C010
GND
PAL2
PROG_A13= PAGED ? A15 : A13
PROG_A14= PAGED ? A16 : A14
PROG_A15= PAGED ? A17 : A15
PROG_A16= PAGED ? A18 : 0
PROGCS1 = A19 & PAGED
PROGCS2 = UNPAGED & (A19 | PAGED)
PAL16L8
053887
Floating (no driver)
verified on the PCB
while running the game
Constant 1.8V observed
MBRIT[2..0]
MBRIT0
MBRIT1
MBRIT2
RMSCRRD
WD_OUT1
WD_OUT2
RMSCRRD
OUTCS
R/W
PAL3
053888
BANKR
BANK1
WRK
WORKCS
OBJRAM_CS
PALETTECS
i7
i6
PAL3
o12 = i7 = 2000-2FFF when BANK1
2800-5FFF
6000-7FFF when BANKR
o13 = i6 = 3C00-3FFF
PAGED= 6000-7FFF when BANKR
8000-9FFF
PALETTE_CS= 0000-07FF when WRK
OBJRAM_CS= 2000-27FF when BANK1
WORKCS= 0000-07FF when WRK
0800-1FFF
UNPAGED= A000-FFFF (ROM PPAL)
PAL1
053884
FMCS
VCS
OBJSET
PCU_CS
MISC
IOCS
JOYSTK
PAL1
JOYSTK 3F8C-3F8F
IO_CS 3F90-3F9F
MISC 3FC0-3FCF
PCU_CS 3FB0-3FBF
OBJSET 3FA0-3FAF
VCS= AS&i7&(A8&A7&A9&AB+i6+A7)
74LS139
decoder
Connected to pin
17 of KONAMI 053260
(D8) in Parodius Da!
E17
CMOS Static RAM
8K x 8 BITS
MB8464A-10W-15W
TEST HEADER
FOOTPRINT MAIN CPU
Conn_02x25_Counter_Clockwise
FP2
FP1
VSS
JOTEGO
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JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Surprise Attack
Size: A3
Date: 2025-09-01
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 2/10



JOTEGO
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JOTEGO
Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 3/10

NOT POPULATED

RA8 4kX8 · RA13 4kX8 · RA12 4kX8 · RA10 4kX8

GFXROMD0 GFXROMD1 GFXROMD2 GFXROMD3 GFXROMD4 GFXROMD5 GFXROMD6 GFXROMD7
GFXROMD8 GFXROMD9 GFXROMD10 GFXROMD11 GFXROMD12 GFXROMD13 GFXROMD14 GFXROMD15
GFXROMD16 GFXROMD17 GFXROMD18 GFXROMD19 GFXROMD20 GFXROMD21 GFXROMD22 GFXROMD23
GFXROMD24 GFXROMD25 GFXROMD26 GFXROMD27 GFXROMD28 GFXROMD29 GFXROMD30 GFXROMD31

GFXROMA[17..0]

CN7 — Connector_Generic_Conn_02x32_Odd_Even_First

K13 / K19 — 955D05-06-07-08 KONAMI MASK 4M - 16BIT — HN62414P — CMOS ROM

GFXROMD[31..0]

J15 KONAMI 051962 TILE DRAWING

DSA[7..0] SCRA

FIX — DFI[7..0]

MAIN_DB[7..0]

DSB[7..0] SCRB

C15C 74LS07 · R18 470Ω · R20 100Ω · CSYNC

JP6 Jp_2_Bridged · JP5 Jp_2_Bridged

G22 XTAL_OSC 24MHz · CP1 104

74LS04 H21A · 74LS04 H21C · H22B 74LS74

WDRST · WDOGRST · WDRSTD

MAIN_A[15..0]

J20 KONAMI 052109 TILE MAP GENERATOR

30 pins per side

LAYER A SCROLL X · LAYER A SCROLL X · LAYER B SCROLL X · TILE ATTRIBUTES · GFX ROM ADDRESS · VRAM ADDRESS · VRAM DATA · CPU DATA · GFX ROM read

SCRAMA[12..0]

H24 · J24 — MB8464A-10W-15W CMOS Static RAM 8K x 8 BITS

VD[15..0]

MAIN_DB[7..0]

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Sheet: /scroll/
File: scroll.kicad_sch

Title: Surprise Attack

Size: A3 | Date: 2025-09-01 | Rev: José Tejada G.
KiCad E.D.A. 8.0.5 | Id: 4/10

NOT POPULATED

MAIN_A[10..0]

OBJROMA[18..0]

OBJROMD[31..0]

ORAMA[10..0]

OBJROMD[7..0]

MAIN_DB[7..0]

HP[9..0]

AR[2..0]

PL[6..0]

BR[2..0]

OBJPXL[8..0]

VCC

VSS

U41
KONAMI
053245
OBJECT GENERATOR

U42
KONAMI
053244
OBJECT DRAWING

HM6116P-3

74LS245

ROM's MASK OF THE SPRITES

K2 955D05-06-07-08 KONAMI MASK 4M - 16BIT

K8 955D05-06-07-08 KONAMI MASK 4M - 16BIT

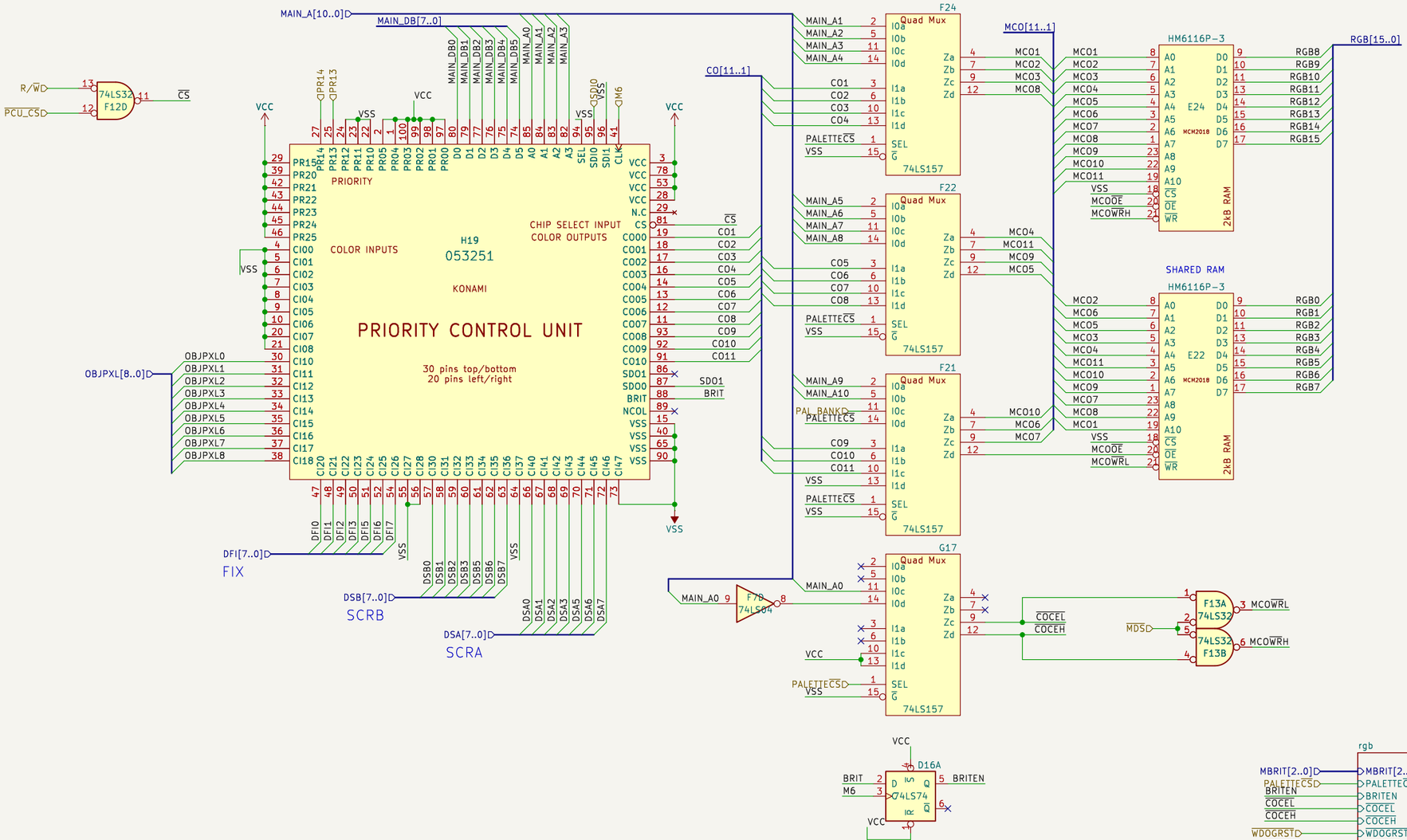
HN62414P

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JOTEGO

Sheet: /objects/
File: objects.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	Rev: José Tejada
KiCad E.D.A. 8.0.5		Id: 5/10

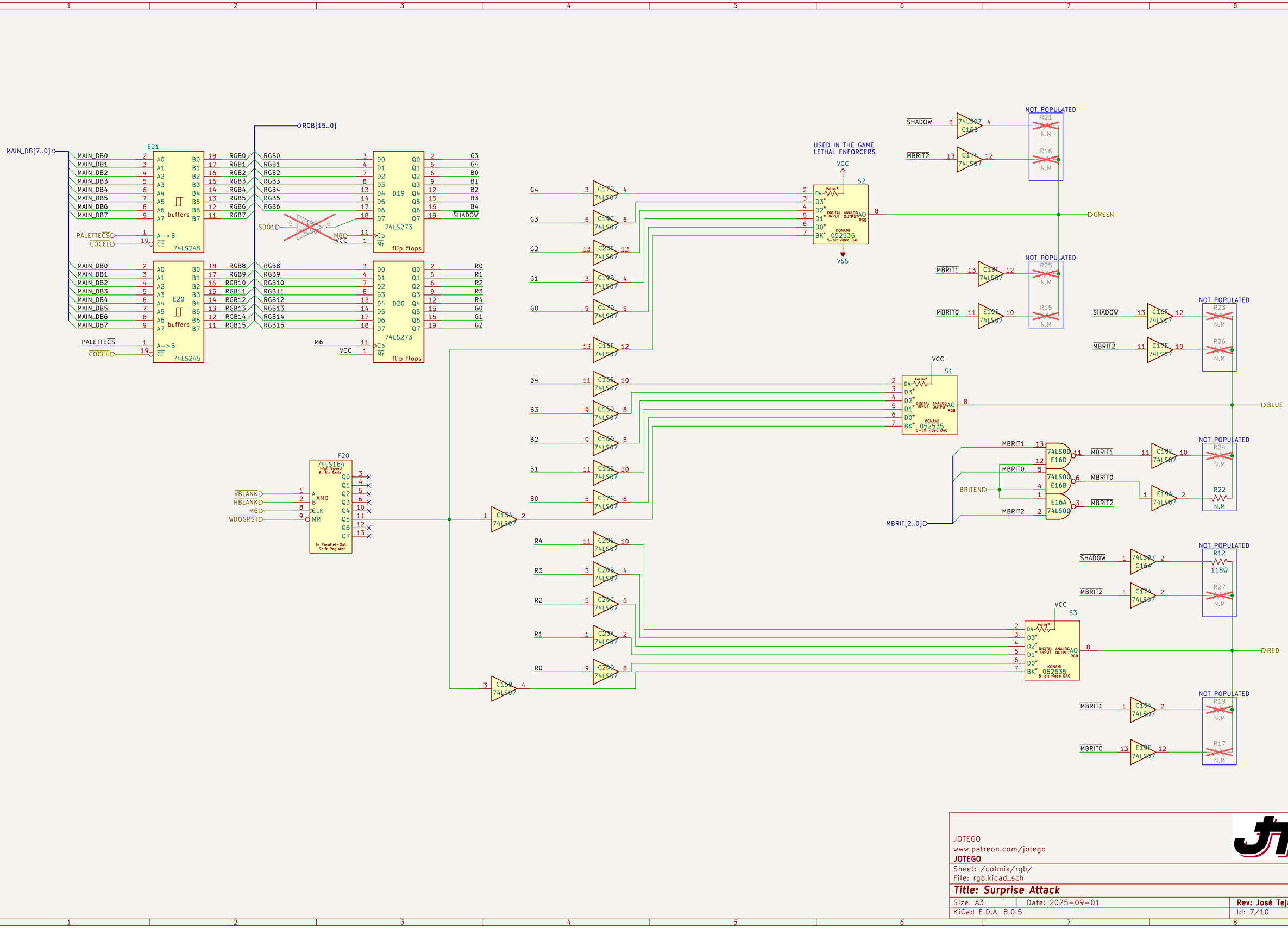


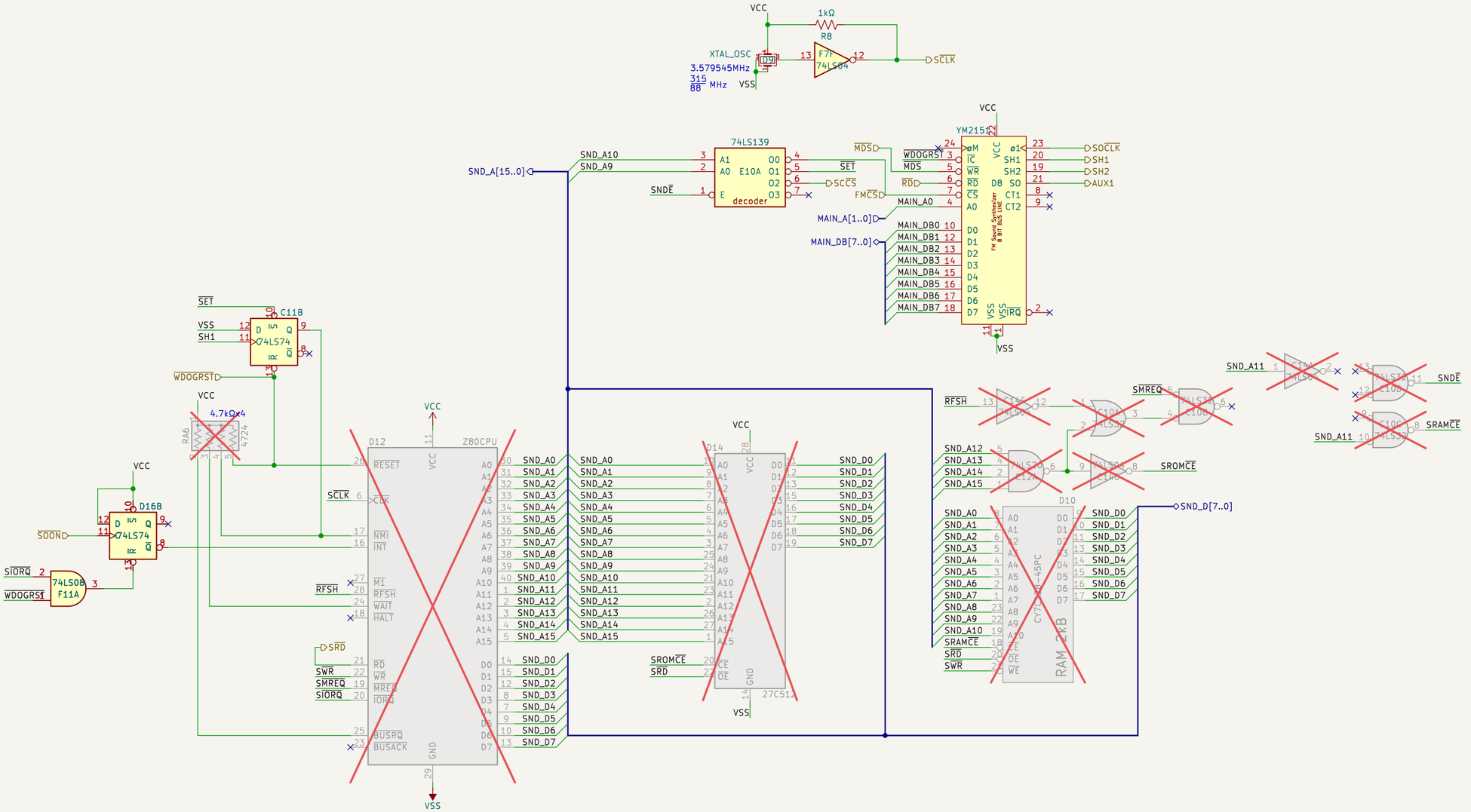
JOTEGO
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JOTEGO

Sheet: /colmix/
File: colmix.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 6/10


RGB[15..0]
MAIN_DB[7..0]
E21
E20
buffers
74LS245
PALETTECS
COCELD
COCEH
74LS273
flip flops
SDO1D
M6
VCC
F20
74LS164
High Speed 8-Bit Serial In Parallel-Out Shift Register
VBLANK
HBLANK
WDOGRST
USED IN THE GAME LETHAL ENFORCERS
S2
S1
S3
KONAMI 052535 5-bit video DAC
DIGITAL INPUT ANALOG OUTPUT
GREEN
BLUE
RED
VCC
VSS
SHADOW
MBRIT2
MBRIT1
MBRIT0
BRITEN
MBRIT[2..0]
NOT POPULATED
N.M
R21
R16
R25
R15
R23
R26
R24
R22
R12
118Ω
R27
R19
R17
74LS00
74LS07
JOTEGO
www.patreon.com/jotego
JOTEGO
Sheet: /colmix/rgb/
File: rgb.kicad_sch
Title: Surprise Attack
Size: A3
Date: 2025-09-01
Rev: José Tejada G.
KiCad E.D.A. 8.0.5
Id: 7/10



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Sheet: /sound/
File: sound.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 8/10

NOT POPULATED

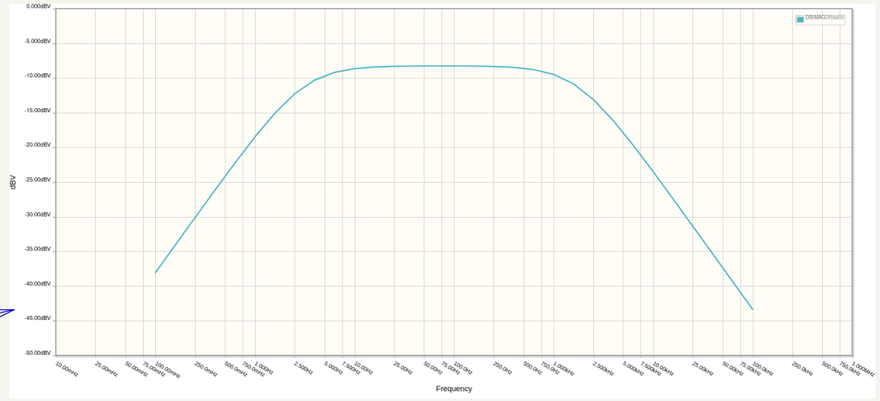
RA[19..0]

Communication between the main CPU
and the audio CPU.

RD[7..0]

Connected to pin 21 of alternate YM2151 footprint.

Connected to pin
7 of 74LS139 (E10)
in Parodius Da!

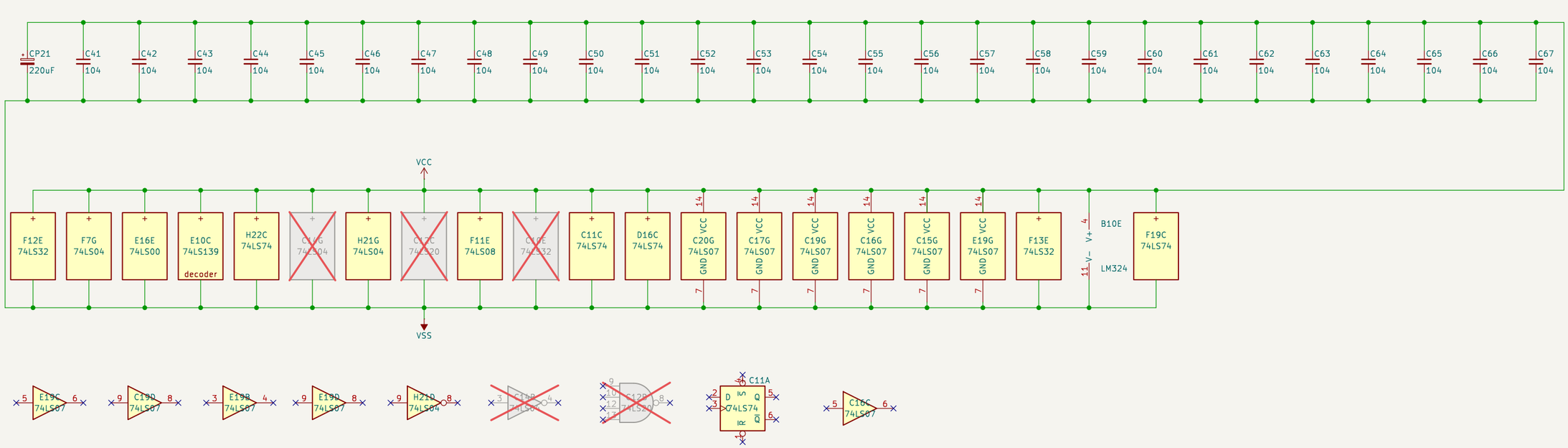


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JOTEGO

Sheet: /pcm/
File: pcm.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 9/10



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Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Surprise Attack

Size: A3	Date: 2025-09-01	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.5		Id: 10/10